Tasks for Joshi: contributing without coding

KERAAL rehabilitation exercise error study

Due Tuesday 13 October 2026 · Questions: ask the team lead

Summary. Task 1 (watching videos) matters most. If time allows, do Task 2 (draft an email). Task 3 (finding papers) is optional. None of them needs coding, and every result goes straight into the study and the paper.

Task	Time	Priority	Deliverable
1. Decode B1-B3 from Group 3 videos	2-3 h	Must	Observation table
2. Draft an email to the dataset authors	~1 h	Should	English email draft
3. Table of inter-rater reliability papers	2-3 h	Optional	Table of 5-8 papers

Background in two minutes

- KERAAL is a public dataset of low-back-pain rehabilitation exercises recorded with a Kinect camera. There are three exercises (**RTK** torso rotation, **ELK** flank stretch, **CTK** hiding face), each with three defined errors (E1-E3).
- Group 3 contains three healthy people (P1-P3) who **acted out the errors on purpose**. The file name says which error.
- What **B1, B2, B3** in the file names mean is not documented anywhere. It could be left/right/both arms, or how strong the error is. Task 1 is to find out by eye.

Error definitions (as published)

Exercise	E1	E2	E3
RTK torso rotation	Arms are not raised enough	The torso's rotation is not sufficient	The body is leaned on the side
ELK flank stretch	Opposite arm is not along the body	Body is not tilted	The above arm is not bent
CTK hiding face	Arms are not raised enough	Arms are not outspread enough	Arms are not raised enough (same wording as E1)

CTK E1 and E3 have the same published wording. Please note how they actually differ in the videos.

Task 1. Decode B1-B3 from Group 3 videos (must)

- **You get**: 36 videos (about 570 MB) from the team lead by USB or shared folder, for participant P1 only: 3 exercises × (3 correct + 9 error videos).
- **Reading a file name**: G3-Anon-**CTK**-P1**T1**-Unknown-**E1B2**-0.avi means exercise CTK, trial 1, error E1, variant B2. **C** means correct.
- **Steps**: (1) pick an exercise; (2) watch the correct (C) video first to learn the right movement; (3) watch B1, B2 and B3 of the same error back to back and look for the difference; (4) fill in the table; (5) repeat for the other errors and exercises.
- **Look for**: which side (left, right or both, **from the person's point of view, not the screen's**), how strong (slight or large), and when during the movement the error happens.
- If you are unsure, write "unsure". That is far more useful than a guess.

Observation table (please fill in)

Exercise	Error	What you see in B1	What you see in B2	What you see in B3	Confidence
RTK	E1				high / mid / low
RTK	E2				high / mid / low
RTK	E3				high / mid / low
ELK	E1				high / mid / low
ELK	E2				high / mid / low
ELK	E3				high / mid / low
CTK	E1				high / mid / low
CTK	E2				high / mid / low
CTK	E3				high / mid / low

Finish with a 2-3 line conclusion, for example: "B1 = left, B2 = right, B3 = both. CTK E3 differs from E1 in ..."

Video list (participant P1, 36 files)

Exercise	Correct	Error videos
RTK	P1T1-C-0, P1T2-C-0, P1T3-C-0	P1T1: E1B1, E1B2, E1B3 · P1T2: E2B1, E2B2, E2B3 · P1T3: E3B1, E3B2, E3B3
ELK	P1T1-C-0, P1T2-C-0, P1T3-C-0	P1T1: E1B1, E1B2, E1B3 · P1T2: E2B1, E2B2, E2B3 · P1T3: E3B1, E3B2, E3B3
CTK	P1T1-C-0, P1T2-C-0, P1T3-C-0	P1T1: E1B1, E1B2, E1B3 · P1T2: E2B1, E2B2, E2B3 · P1T3: E3B1, E3B2, E3B3

Full name example: G3-Anon-RTK-P1T2-Unknown-E2B3-0.avi

Done when: all 9 rows are filled ("unsure" is allowed) and the 2-3 line conclusion is written.

Task 2. Draft an email to the dataset authors (should)

- **Why**: four questions cannot be answered from the data alone.
- **Do**: write a polite English **draft** that asks the questions below. A translator is fine. Add one or two sentences introducing yourself and the course project.
- **To**: Sao Mai Nguyen (nguyensmai@gmail.com) and Maxime Devanne (maxime.devanne@uha.fr), the contacts listed in the dataset's format readme.
- **Important**: **do not send it yourself.** Give the draft to the team lead; it will be sent after review.

#	Question to include
1	Is there a table that links Group 1B recordings to patients and sessions? The file names contain no patient ID.
2	Are participant S1 at Brest and participant S1 at Roscoff in Group 2A the same person?
3	What do the codes B1-B3 in Group 3 file names mean? (Mention what you observed in Task 1.)
4	Two annotator-B files have a corrupted end time ("tart:"): G2A-RTK-S1-Roscoff-050 and G2A-RTK-S6-Roscoff-077. Is a corrected version available?

Done when: one draft with a greeting, a short introduction, the four questions and a thank-you.

Task 3. Table of inter-rater reliability papers (optional)

- **Why**: our key finding is that the two experts apply different criteria to the same movement. Evidence that this is common in physiotherapy assessment makes the paper's argument stronger.
- **Do**: find 5-8 papers on Google Scholar and fill in the table below. Reading the abstract is enough.
- **Search terms**: "inter-rater reliability physiotherapy exercise assessment", "movement quality rating agreement", "observational movement assessment reliability kappa", "low back pain exercise technique rating".

Title, year	Link	Movement rated	Raters	Agreement (κ, ICC)	One-line summary

Done when: at least 5 papers, preferably ones that report agreement numbers.

Please keep in mind

- The videos and data are for research only (licence CC BY-NC-SA). Do not post or forward them, and delete your copy when you are done.
- Do not try to find out who the people in the videos are. We only look at the movement.
- If you get stuck, ask right away. A one-line status update on Tuesday morning is enough.
- Send the deliverables (observation table, email draft, paper table) to the team lead by Tuesday, in any format (Word, Google Docs, Hangul).